

ULIP Integration Requirement Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	15/06/2023	Ayush Arya	Nayab Ali	Baseline Version
2.0	01/10/2024	Ayush Arya	Nayab Ali	Updated
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for FCI system information from ULIP system.

This document details the following points:

1. **Data Integration between FCI and ULIP web service:** Identification of operational points where data integration between **FCI** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will just hit the API and get the data (no need to share any data in the request body). ULIP system will call respective **FCI** API for accessing information.

Following are the points for accessing vehicle information: -

1. This API will share the vehicle information data to the user.
2. ULIP will expose API to get data from FCI API.
3. **FCI/01** API will take no data in request body from user and get data from FCI API.

1.2 Data Integration between FCI System and ULIP application

Users will share information to ULIP system based on data shared ULIP will provide data to user.

1.3 FCI/01

This API will connect with FCI API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	N/A	N/A	N/A	N/A	N/A

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide detail is required.
3. FCI API will share details of FCI vehicle information which exist in FCI system in JSON format.
4. FCI Integration will add JSON in **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

Example: Data is shared by ULIP system, then FCI-integration will make a request to FCI API.

Request: The request must be of the following format and passed within the URL body in JSON format.

For staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FCI/01>

```
{}
```

Curl URL

```
curl --location --request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FCI/01' \  
-- \  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNjg3MzM2NDk4LCJhcHBzIjoZGF0YXB1c2gifQ.r0kHctO0lmLe-taVNp_u-  
tV2UJebPHk48scHdhJIZgk_n54sBu3N9vXYiEFTmMwfERn9aHv2Qo9fHPugUgg48w' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{}'
```

For production: <https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FCI/01>

```
{}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FCI/01' \  
--header'authorization:Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI3NzYyNzcwLCJhcHBzIjoZGF0YXB1c2gifQ.9_kJwxuXfJYoMt0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9oZ_B4hiiq2ZCg-  
yAbIUgL6YN-TVie3cA' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data '{}'
```

For obtaining authentication token, refer [Section 1.4](#)

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Response

To acknowledge the above request, ULIP will send the below response -

1. Success response

```
{  "response": [
    {
      "response": {
        "trips": [
          {
            "VehicleRegNo": "RJ09GB0797",
            "Latitude": "24.192329",
            "Longitude": "82.770493",
            "ServerlogDateTime": "2023-06-15 09:08:23",
            "Commodity": "Wheat",
            "TrackingDateTime": "2023-06-15 09:08:05"
          },
          {
            "VehicleRegNo": "HR576733",
            "Latitude": "29.41037",
            "Longitude": "74.212463",
            "ServerlogDateTime": "2023-06-15 09:35:42",
            "Commodity": "Wheat",
            "TrackingDateTime": "2023-06-15 09:35:25"
          },
          {
            "VehicleRegNo": "HR57A5033",
            "Latitude": "29.407866",
            "Longitude": "74.220901",
            "ServerlogDateTime": "2023-06-15 09:35:44",
            "Commodity": "Wheat",
            "TrackingDateTime": "2023-06-15 09:35:26"
          },
          {
            "VehicleRegNo": "RJ31GA9250",
            "Latitude": "29.482855",
            "Longitude": "74.68367",
            "ServerlogDateTime": "2023-06-15 09:35:41",
            "Commodity": "Wheat",
            "TrackingDateTime": "2023-06-15 09:35:24"
          },
          {
            "VehicleRegNo": "RJ41GA1433",
            "Latitude": "29.491783",
            "Longitude": "74.71566",
```

```

        "ServerlogDateTime": "2023-06-15 09:34:16",
        "Commodity": "Wheat",
        "TrackingDateTime": "2023-06-15 09:33:58"
    },
    {
        "VehicleRegNo": "RJ52GA5444",
        "Latitude": "25.326105",
        "Longitude": "74.634727",
        "ServerlogDateTime": "2023-05-30 22:55:35",
        "Commodity": "Wheat",
        "TrackingDateTime": "2023-05-30 22:55:24"
    }
],
"message": {
    "IsVerified": "Y",
    "MessageInfo": "Success",
    "Status": "S",
    "MobileNo": null,
    "Email": null,
    "refreshtoken": null,
    "token": null,
    "DesignationId": null
}
},
"responseStatus": "SUCCESS"
}
],
"error": "false",
"code": "200",
"message": "Success"}

```

Status Code 200: FCI data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: Server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in FCI system through a data exchange URL that has following form:

For staging: <https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FCI/01>

```
{}
```

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Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Summary of HTTP Basic Authentication is as follows:

First time User hits ULIP secured API through username and password for access token as follow-

```
curl --location --request POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{"username": "xxxx", "password": "xxxx@123"}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiI1bGlwliwiaWF0IjoxNjg3MzM2NDA1LCJhcHBzIjoiaZGF0YXB1c2gifQ.r0kHctO0lmLe-taVNp_u-
tV2UJebPHk48scHdhJIZgk_n54sBu3N9vXYiEFTmMwfERn9aHv2Qo9fHPugUgg48w'
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represents user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

```
curl --location --request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FCI/01' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiI1bGlwliwiaWF0IjoxNjg3MzM2NDA1LCJhcHBzIjoiaZGF0YXB1c2gifQ.r0kHctO0lmLe-taVNp_u-
tV2UJebPHk48scHdhJIZgk_n54sBu3N9vXYiEFTmMwfERn9aHv2Qo9fHPugUgg48w' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{}'
```

For production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{ "username": "xxxx", "password": "xxxx@123" }'
```


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Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represents user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FCI/01>

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FCI/01' \  
--header 'authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.9_kJwxuXfJYoMt0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9olemZ_B4hiiq2ZCog-yAbIUgL6YN-TVie3cA' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data '{}'
```